

CytoHybrid-OS: Hybrid Virtual Tissue Software: A Transcriptomics-Informed Virtual Immune-Tissue Model for Benchmarking Hybrid Mechanistic–AI Architectures, Including Residual Correction, AI-Parameterized Mechanistic Prediction, Neural-ODE-Like Dynamics, and Reduced PINNs

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Proof-of-concept

Abstract

Inflammatory immune responses are controlled by nonlinear feedback among tissue damage, immune activation, cytokine amplification, regulatory resolution, cell–cell communication, spatial exclusion, and repair. Standard machine-learning models trained on bulk transcriptomic features can classify broad inflammatory states but often fail to recover latent biological control mechanisms or predict responses to interventions. We developed a transcriptomics-informed virtual immune-tissue modeling framework that couples an agent-based model of mobile immune cells carrying internal gene-program states with smooth spatial latent fields representing tissue damage, immune activation, cytokine signaling, and resolution. The simulator generates pseudo-bulk RNA-seq-like features, spatial transcriptomic module maps, mechanistic summaries, hidden parameters, and intervention counterfactuals. The updated computational benchmark compares a broad non-Bayesian family of hybrid mechanistic–AI architectures: pure ML baselines, mechanistic summaries, Type I feature-level hybrid integration, Type II mechanistic prediction with AI residual/error correction, Type III AI-inferred mechanistic parameterization, Type III+ AI-parameterized mechanistic prediction with residual correction, Type IV reduced state-space physics-informed neural network (PINN), Type V Neural-ODE/universal-differential-equation-like learned residual dynamics, Type VII simulator surrogate/emulator diagnostics, zero-shot drug-response prediction, and one-/two-shot drug-response adaptation. The framework is designed as a simulation-to-real bridge: simulated RNA module scores can later be replaced by real bulk RNA-seq, single-cell RNA-seq, spatial transcriptomics, imaging, and longitudinal clinical response data. The long-term goal is patient-specific digital-twin modeling for precision inflammation and immunotherapy.

1 Purpose and scientific motivation

The purpose of this study is to construct a controlled virtual immune-tissue system in which the true hidden biology is known, generate realistic transcriptomic and spatial readouts, and test which modeling architectures can recover immune-inflammatory regimes, predict future inflammatory burden, infer latent parameters, and forecast intervention response. The model is not intended to claim that simulated tissues are real tissues. Instead, it provides a benchmark in which known causal mechanisms are embedded and alternative AI, mechanistic, and hybrid models can be compared under controlled conditions.

The biological premise is that inflammation is not only a molecular phenotype but also a dynamical and spatial process. Bulk RNA-seq captures average molecular state. Spatial transcriptomics captures tissue architecture, such as immune exclusion and core-rim mismatch. Mechanistic models encode feedback laws and intervention mechanisms. AI/ML can infer latent structure, learn residual errors, emulate expensive simulators, and adapt to early drug-response data. A rigorous hybrid framework should therefore compare multiple forms of mechanistic-AI integration rather than only concatenating mechanistic summaries with omics features.

2 Hypotheses tested

The simulation benchmark tests the following hypotheses.

- H1. **Pseudo-bulk limitation hypothesis.** Pseudo-bulk RNA-like module features can capture overall inflammatory intensity but lose spatial organization and therefore perform poorly for immune-exclusion and latent-parameter recovery.
- H2. **Spatial added-value hypothesis.** Spatial transcriptomic features improve classification of inflammatory regimes and inference of spatial mechanisms, especially barrier strength, cytotoxic-damage colocalization, and immune exclusion.
- H3. **Mechanistic-information hypothesis.** Mechanistic summaries of damage, cytokine, activation, suppression, cytotoxicity, repair, and exhaustion improve future-burden prediction because they preserve dynamical information that cross-sectional omics averages discard.
- H4. **Residual-correction hypothesis.** A mechanistic model captures dominant biology but leaves systematic errors; AI residual correction can improve prediction without discarding interpretability.
- H5. **AI-parameterization hypothesis.** Hidden mechanistic parameters such as inflammatory gain, TGF- β gain, exhaustion gain, repair rate, diffusion, and barrier strength can be inferred from observed RNA/spatial features and used for patient-specific mechanistic prediction.
- H6. **Neural-dynamics hypothesis.** Learned residual vector fields, in Neural-ODE/universal-differential-equation form, can improve prediction when the known mechanistic equations are incomplete.
- H7. **PINN hypothesis.** A physics-informed neural network that predicts reduced immune-state trajectories while penalizing violations of immune-dynamical equations can provide a stronger data-efficient hybrid model than unconstrained ML.
- H8. **Few-shot drug-learning hypothesis.** A model trained only on untreated states may be insufficient for drug response; one or two labeled intervention examples per drug can improve prediction by learning drug-specific residual corrections.

3 Overview of the complete computational pipeline

For each virtual subject, the workflow is:

1. Select an inflammatory regime.
2. Sample subject-specific mechanistic parameters.

3. Simulate a transcriptomics-informed immune-tissue ABM.
4. Generate smooth latent tissue fields and RNA-seq-like spatial module maps.
5. Extract pseudo-bulk, spatial, mechanistic, and RNA-like features.
6. Compute future untreated inflammatory burden and hidden mechanistic parameters.
7. Simulate intervention counterfactuals and identify the best intervention.
8. Split virtual subjects into training and testing cohorts.
9. Compare pure ML, mechanistic, and multiple hybrid MM–AI models.
10. Run zero-shot and few-shot drug-response prediction experiments.

4 Inflammatory regimes

The virtual cohort contains five regimes:

1. **Acute resolving:** early damage and inflammation followed by resolution and repair.
2. **Chronic:** persistent injury and incomplete resolution.
3. **Hyperinflammatory:** strong cytokine-positive feedback and high inflammatory burden.
4. **Immune excluded:** damaged core with stromal/barrier-mediated exclusion of cytotoxic immune activity.
5. **Suppressed:** weak effector activation with dominant regulatory, suppressive, or exhaustion-like programs.

Each regime is defined by a different distribution over mechanistic parameters, not merely by a label. Thus, classification requires recovering underlying biological dynamics rather than memorizing labels.

5 Spatial latent tissue-field model

Although the ABM contains individual cells, the upgraded code also generates smooth latent fields to make the spatial readouts resemble tissue imaging and spatial transcriptomics. At spatial position $(x, y) \in \Omega \subset [0, 1]^2$ and time t , the reduced tissue state is

$$\mathbf{z}(x, y, t) = (D(x, y, t), I(x, y, t), C(x, y, t), R(x, y, t)), \quad (1)$$

where D is damage/stress, I is immune activation, C is cytokine/inflammatory signaling, and R is regulatory/resolution activity.

5.1 Spatial templates

An injury template is defined by a smooth hotspot:

$$J(x, y) = \exp \left[-\frac{(x - x_J)^2 + (y - y_J)^2}{2r_J^2} \right]. \quad (2)$$

A barrier field $B(x, y)$ represents extracellular matrix, stromal exclusion, or suppressive tissue architecture. In practice, the code builds this from core and rim Gaussian-like fields and normalizes it to $[0, 1]$. Immune access $A(x, y)$ decreases with distance from the immune source and is further reduced by barrier strength:

$$A(x, y) = A_0(x, y)\{1 - bB(x, y)\}, \quad (3)$$

where b is the subject-level barrier parameter.

5.2 Field equations

The conceptual field dynamics are

$$\frac{\partial D}{\partial t} = S_D(x, y, t) + P_D(x, y) - k_D D - k_{\text{repair}} R D, \quad (4)$$

$$\frac{\partial I}{\partial t} = \alpha_D D A(x, y) \Pi_B(x, y) + \alpha_C C \Pi_B(x, y) - \delta_I I - \gamma_R R I, \quad (5)$$

$$\frac{\partial C}{\partial t} = \beta_I I + \beta_D D - \delta_C C - \eta_R R C + \mathcal{D}_C \nabla^2 C, \quad (6)$$

$$\frac{\partial R}{\partial t} = \rho_C \frac{C^n}{K_C^n + C^n} - \delta_R R. \quad (7)$$

Here

$$S_D(x, y, t) = a_J \exp \left[-\frac{(t - t_J)^2}{2\sigma_J^2} \right] J(x, y) \quad (8)$$

is acute injury input, P_D is persistent injury, and

$$\Pi_B(x, y) = 1 - \lambda_B b B(x, y) \quad (9)$$

penalizes immune activation in barrier-rich regions.

Biologically, the feedback structure is

$$D \rightarrow I \rightarrow C \rightarrow R \dashv (D, I, C). \quad (10)$$

Damage activates immunity, immune activation produces cytokines, cytokines induce resolution, and resolution suppresses damage, immune activation, and cytokines. Spatial barriers modulate where immune activation can occur.

6 Transcriptomics-informed agent-based cell model

Each virtual immune cell i has position

$$\mathbf{x}_i(t) = (x_i(t), y_i(t)) \quad (11)$$

and cell type $q_i \in \{\text{macrophage, T cell, NK cell, Treg}\}$. Each cell carries an internal transcriptomic program vector

$$\mathbf{g}_i(t) = \left(g_i^{\text{NF}\kappa\text{B}}, g_i^{\text{STAT3}}, g_i^{\text{IFNG}}, g_i^{\text{cytotoxic}}, g_i^{\text{exhaustion}}, g_i^{\text{TGF}\beta}, g_i^{\text{repair}}, g_i^{\text{migration}} \right). \quad (12)$$

These programs are not merely downstream readouts; they regulate cell behavior.

6.1 Gene-program dynamics

Each program relaxes toward a signal-dependent target:

$$\frac{dg_{ik}}{dt} = \lambda_k [G_k(D_i, C_i, B_i, q_i, \text{age}_i) - g_{ik}], \quad (13)$$

where $D_i = D(\mathbf{x}_i, t)$, $C_i = C(\mathbf{x}_i, t)$, and $B_i = B(\mathbf{x}_i)$. Examples include

$$G_{\text{NF}\kappa\text{B}} = \text{sigmoid}(a_D D_i + a_C C_i - \theta_N) g_{\text{NF}\kappa\text{B}, \text{gain}}, \quad (14)$$

$$G_{\text{STAT3}} = \text{sigmoid}(a_C C_i + a_D D_i - \theta_S) g_{\text{STAT3}, \text{gain}}, \quad (15)$$

$$G_{\text{TGF}\beta} = \text{sigmoid}(a_C C_i + a_B B_i + a_D D_i - \theta_T) g_{\text{TGF}\beta, \text{gain}}, \quad (16)$$

$$G_{\text{exhaustion}} = \text{sigmoid}(a_C C_i + a_B B_i + a_{\text{age}} \text{age}_i / \text{maxage}_i - \theta_E) g_{\text{exhaustion}, \text{gain}}, \quad (17)$$

$$G_{\text{cytotoxic}} = \text{sigmoid}(a_D D_i + a_C C_i - \theta_K) g_{\text{cytotoxic}, \text{gain}}. \quad (18)$$

6.2 Cell activation, secretion, suppression, cytotoxicity, and repair

Cell activation is represented as

$$A_i = \text{clip} \left(w_N g_i^{\text{NF}\kappa\text{B}} + w_S g_i^{\text{STAT3}} + w_F g_i^{\text{IFNG}} - w_E g_i^{\text{exhaustion}} - w_T g_i^{\text{TGF}\beta}, 0, 1 \right). \quad (19)$$

Macrophage cytokine secretion follows

$$s_i^C = \left[a_N g_i^{\text{NF}\kappa\text{B}} + a_S g_i^{\text{STAT3}} - a_R g_i^{\text{repair}} \right]_+. \quad (20)$$

T/NK cytotoxicity follows

$$K_i = g_i^{\text{cytotoxic}} \left(1 - q_E g_i^{\text{exhaustion}} \right). \quad (21)$$

Treg suppression is approximated by

$$S_i = a_T g_i^{\text{TGF}\beta} + a_E g_i^{\text{exhaustion}}. \quad (22)$$

Repair activity is

$$Q_i = a_R g_i^{\text{repair}} + a_T g_i^{\text{TGF}\beta}. \quad (23)$$

6.3 Cell walking and communication

Cells move by chemotaxis and random walking:

$$\frac{d\mathbf{x}_i}{dt} = m_i \left[w_i \frac{\nabla C(\mathbf{x}_i, t)}{\|\nabla C(\mathbf{x}_i, t)\| + \epsilon} + \sigma \boldsymbol{\xi}_i(t) \right], \quad (24)$$

where motility depends on migration, barrier, and exhaustion:

$$m_i = m_0 + m_1 g_i^{\text{migration}} (1 - q_B B_i) - m_2 g_i^{\text{exhaustion}}. \quad (25)$$

When two cells come within a contact radius r_c , they communicate directly. Tregs suppress nearby inflammatory and cytotoxic programs:

$$g_j^{\text{NF}\kappa\text{B}} \leftarrow g_j^{\text{NF}\kappa\text{B}} - \epsilon S_i, \quad (26)$$

$$g_j^{\text{IFNG}} \leftarrow g_j^{\text{IFNG}} - \epsilon S_i, \quad (27)$$

$$g_j^{\text{cytotoxic}} \leftarrow g_j^{\text{cytotoxic}} - \epsilon S_i, \quad (28)$$

$$g_j^{\text{exhaustion}} \leftarrow g_j^{\text{exhaustion}} + \epsilon S_i. \quad (29)$$

Macrophages can activate nearby T/NK cells, while T/NK-derived IFN-like signals amplify macrophage inflammatory programs. In addition to contact communication, cells communicate indirectly through cytokine fields.

7 RNA-seq-like and spatial transcriptomic observation model

The simulator generates module maps $M_m(x, y, t)$ from hidden fields and cell programs:

$$M_m(x, y, t) = f_m(D, I, C, R, B, A, \mathbf{g}) + \epsilon_m(x, y, t). \quad (30)$$

Examples are

$$M_{\text{TNF/NF}\kappa\text{B}} = 0.58C + 0.32I + 0.10D + \epsilon, \quad (31)$$

$$M_{\text{IL6/STAT3}} = 0.70C + 0.18D + 0.12I + \epsilon, \quad (32)$$

$$M_{\text{IFNG}} = 0.65I + 0.25C + 0.10A + \epsilon, \quad (33)$$

$$M_{\text{cytotoxic}} = 0.65IA + 0.18C + \epsilon, \quad (34)$$

$$M_{\text{TGF}\beta} = 0.55R + 0.30B + 0.12D + \epsilon, \quad (35)$$

$$M_{\text{EMT/fibrosis}} = 0.55B + 0.30R + 0.10D + \epsilon, \quad (36)$$

$$M_{\text{damage}} = 0.78D + 0.16C + \epsilon. \quad (37)$$

Pseudo-bulk features integrate over tissue:

$$\bar{M}_m = \frac{1}{|\Omega|} \int_{\Omega} M_m(x, y, T) \, dx dy. \quad (38)$$

Spatial features include module variance, hotspots, core–outside contrasts, hotspot distances, colocalization, and immune-exclusion indices. One important immune-exclusion feature is

$$E_{\text{excl}} = (\bar{M}_{\text{cytotoxic, outside}} - \bar{M}_{\text{cytotoxic, core}}) \bar{M}_{\text{damage, core}}. \quad (39)$$

8 Outcomes

The main future burden outcome is

$$Y_{\text{burden}} = \bar{C}(T) + \bar{I}(T) + \frac{1}{2}\bar{D}(T). \quad (40)$$

For intervention $a \in \mathcal{A}$, the treated burden is

$$Y_{\text{burden}}^{(a)} = \bar{C}^{(a)}(T) + \bar{I}^{(a)}(T) + \frac{1}{2}\bar{D}^{(a)}(T). \quad (41)$$

The optimal intervention label is

$$a^* = \arg \min_{a \in \mathcal{A}} Y_{\text{burden}}^{(a)}. \quad (42)$$

9 Intervention mechanics

Candidate interventions alter mechanistic parameters.

- **Anti-cytokine:** increases cytokine decay and reduces cytokine amplification, e.g. $\delta_C \uparrow$, $\beta_I \downarrow$, $\alpha_C \downarrow$.
- **Pro-resolution:** strengthens resolution and repair, e.g. $\rho_C \uparrow$, $\gamma_R \uparrow$, $\eta_R \uparrow$, $k_{\text{repair}} \uparrow$.
- **Barrier modulation:** reduces barrier strength $b \downarrow$, improving immune access.

- **Anti-exhaustion:** decreases exhaustion gain and increases cytotoxic/IFN-like activity in the ABM layer.
- **Combination therapy:** modifies cytokine, resolution, repair, barrier, and exhaustion-related parameters simultaneously.

10 Model classes tested in the Python code

The updated code compares pure ML, mechanistic, and multiple hybrid models. Bayesian methods are deliberately excluded.

10.1 Pure ML baselines

Pure ML baselines include pseudo-bulk marker modules, RNAseq-like gene features, and bulk+spatial marker features:

$$\hat{Y} = f_{ML}(X_{\text{bulk}}), \quad \hat{Y} = f_{ML}(X_{\text{RNA}}), \quad \hat{Y} = f_{ML}(X_{\text{bulk}} \parallel X_{\text{spatial}}). \quad (43)$$

These baselines test the value of observation layers without explicit mechanistic structure.

10.2 Mechanistic-summary model

The mechanistic-summary model uses features derived from simulator state trajectories:

$$X_{\text{mech}} = \{D, I, C, R, \text{activation, repair, suppression, cytotoxicity, AUCs, peaks, time-slices}\}. \quad (44)$$

This is strong in simulation because the hidden state is available. In real data, these quantities would need to be inferred.

10.3 Type I: feature-concatenation hybrid

The simplest hybrid model concatenates observed and mechanistic features:

$$\hat{Y} = f_I(X_{\text{RNA}} \parallel X_{\text{bulk}} \parallel X_{\text{spatial}} \parallel X_{\text{mech}}). \quad (45)$$

This is the original hybrid benchmark. It is simple and interpretable but does not allow AI to alter the mechanistic simulation.

10.4 Type II: mechanistic prediction plus AI residual correction

A mechanistic predictor produces $\hat{Y}_{MM}$. AI learns the residual:

$$\Delta = Y - \hat{Y}_{MM}, \quad \hat{\Delta} = f_{II}(X_{\text{obs}}, X_{\text{mech}}, \hat{Y}_{MM}). \quad (46)$$

The final prediction is

$$\hat{Y}_{II} = \hat{Y}_{MM} + \hat{\Delta}. \quad (47)$$

This architecture is useful when the mechanistic model captures first-order biology but misses unmodeled heterogeneity, nonlinear drug response, or measurement bias.

10.5 Type III: AI-parameterized mechanistic model

In this architecture, AI infers hidden parameters from observable data:

$$\hat{\theta} = f_{\theta}(X_{\text{RNA}}, X_{\text{bulk}}, X_{\text{spatial}}). \quad (48)$$

A mechanistic response-surface surrogate then predicts outcome from inferred parameters:

$$\hat{Y}_{III} = M_{\text{sur}}(\hat{\theta}). \quad (49)$$

This is central for precision medicine because real patients do not reveal their mechanistic parameters directly. The model must infer patient-specific inflammatory gain, barrier strength, repair capacity, cytotoxic gain, exhaustion gain, and regulatory tone from measurable molecular and spatial data.

10.6 Type III+: AI-parameterized MM plus residual correction

This architecture combines Type II and Type III:

$$\hat{\theta} = f_{\theta}(X_{\text{obs}}), \quad (50)$$

$$\hat{Y}_{\text{paramMM}} = M_{\text{sur}}(\hat{\theta}), \quad (51)$$

$$\hat{Y}_{III+} = \hat{Y}_{\text{paramMM}} + f_{\text{resid}}(X_{\text{obs}}, \hat{\theta}, \hat{Y}_{\text{paramMM}}). \quad (52)$$

This is often a practical best compromise: hidden parameters provide interpretability, while residual correction provides flexibility.

10.7 Type IV: reduced state-space PINN

The new Python code includes a lightweight reduced state-space PINN. This is not a full spatial PDE PINN over (x, y, t) ; it is a reduced immune-state PINN over time-slice summaries. The neural network maps baseline observed features to a reduced trajectory:

$$\mathcal{N}_{\phi}(X_{\text{obs}}) = (\hat{z}_{50}, \hat{z}_{75}, \hat{z}_T), \quad (53)$$

where $\mathbf{z} = (D, I, C, R)$ and the subscripts denote intermediate and final time fractions.

The loss combines data fit and mechanistic residual penalties:

$$\mathcal{L}_{PINN} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}}. \quad (54)$$

The data loss is

$$\mathcal{L}_{\text{data}} = \|\mathcal{N}_{\phi}(X_{\text{obs}}) - \mathbf{Z}_{\text{target}}\|^2. \quad (55)$$

The physics loss penalizes violations of the reduced immune equations:

$$\mathcal{L}_{\text{phys}} = \left\| \frac{\hat{z}_{50} - \mathbf{z}_{25}}{\Delta t} - f_{\text{known}}(\mathbf{z}_{25}) \right\|^2 \quad (56)$$

$$+ \left\| \frac{\hat{z}_{75} - \hat{z}_{50}}{\Delta t} - f_{\text{known}}(\hat{z}_{50}) \right\|^2 \quad (57)$$

$$+ \left\| \frac{\hat{z}_T - \hat{z}_{75}}{\Delta t} - f_{\text{known}}(\hat{z}_{75}) \right\|^2. \quad (58)$$

The known reduced vector field used in the code is

$$f_D(\mathbf{z}) = 0.55D(1 - 0.5D) - 0.85RD - 0.08D, \quad (59)$$

$$f_I(\mathbf{z}) = 0.80D + 0.45C - 0.28I - 0.55RI, \quad (60)$$

$$f_C(\mathbf{z}) = 0.75I + 0.30D - 0.40C - 0.45RC, \quad (61)$$

$$f_R(\mathbf{z}) = 0.65 \frac{C^3}{1 + C^3} - 0.18R. \quad (62)$$

The final PINN-predicted burden is

$$\hat{Y}_{PINN} = \hat{C}_T + \hat{I}_T + \frac{1}{2}\hat{D}_T. \quad (63)$$

10.8 Type V: Neural-ODE / universal-differential-equation-like residual dynamics

The new Python code also includes a Neural-ODE-like learned residual dynamics model. It is implemented as a discrete residual-vector-field learner rather than a continuous automatic-differentiation ODE solver. The known reduced dynamics are

$$\frac{d\mathbf{z}}{dt} = f_{known}(\mathbf{z}). \quad (64)$$

The hybrid Neural-ODE-like model assumes

$$\frac{d\mathbf{z}}{dt} = f_{known}(\mathbf{z}) + f_{NN}(\mathbf{z}, X_{\text{obs}}, t; \phi). \quad (65)$$

For training intervals, the empirical derivative is

$$\widehat{\frac{d\mathbf{z}}{dt}} = \frac{\mathbf{z}_{k+1} - \mathbf{z}_k}{\Delta t}. \quad (66)$$

The target residual vector field is

$$\mathbf{r}_k = \frac{\mathbf{z}_{k+1} - \mathbf{z}_k}{\Delta t} - f_{known}(\mathbf{z}_k). \quad (67)$$

An MLP regressor learns

$$\hat{\mathbf{r}}_k = f_{NN}(\mathbf{z}_k, f_{known}(\mathbf{z}_k), t_k, X_{\text{obs}}; \phi). \quad (68)$$

Prediction rolls the state forward:

$$\hat{\mathbf{z}}_{k+1} = \hat{\mathbf{z}}_k + \Delta t [f_{known}(\hat{\mathbf{z}}_k) + f_{NN}(\hat{\mathbf{z}}_k, X_{\text{obs}}, t_k; \phi)]. \quad (69)$$

This is a practical universal-differential-equation approximation: known immune equations provide the backbone, and AI learns missing transition dynamics.

10.9 Type VII: surrogate/emulator diagnostics

The surrogate model trains a fast AI model to approximate simulator outcomes:

$$\hat{Y}_{sur} = f_{sur}(\boldsymbol{\theta}). \quad (70)$$

When true hidden parameters are used, this is an oracle diagnostic and not clinically available. It estimates the upper-bound value of knowing patient-specific mechanistic parameters.

11 Training/testing design

All model comparisons use a held-out virtual-subject test set. Training is performed on simulated subjects generated under the same broad simulator but with subject-specific parameter variation. Tasks include:

1. inflammatory-regime classification;
2. future untreated burden prediction;
3. best-intervention classification;
4. mechanistic-parameter recovery;
5. zero-shot treated-burden prediction;
6. one-/two-shot intervention adaptation.

Regression performance is evaluated using R^2 , RMSE, and MAE. Classification performance is evaluated using accuracy and balanced accuracy.

12 Zero-shot and few-shot drug-response learning

The drug-response workflow asks whether a model trained primarily on untreated dynamics can predict intervention response.

12.1 Zero-shot prediction

A baseline-trained model predicts drug response without labeled examples of that drug-response outcome:

$$\hat{Y}_{zero}^{(a)} = f_0(X_{baseline}, a). \quad (71)$$

12.2 Few-shot residual adaptation

With one or two labeled treated examples per drug, the model learns a residual correction:

$$\Delta^{(a)} = Y^{(a)} - \hat{Y}_{zero}^{(a)}. \quad (72)$$

A small residual model is trained:

$$\hat{\Delta}^{(a)} = h_a(X_{baseline}, X_{early}^{(a)}). \quad (73)$$

The adapted prediction is

$$\hat{Y}_{few-shot}^{(a)} = \hat{Y}_{zero}^{(a)} + \hat{\Delta}^{(a)}. \quad (74)$$

This mimics a pilot-treatment setting in which early molecular/spatial response is observed before final outcome is known.

13 Implementation status of all hybrid models

Model class	Included?	Implementation in updated Python code
Pure ML: pseudo-bulk	Yes	Random-forest models using pseudo-bulk marker/module features.
Pure ML: RNA-like features	Yes	Models using synthetic RNAseq-like feature layer.
Pure ML: spatial transcriptomics	Yes	Bulk plus spatial hotspot, contrast, colocalization, and exclusion features.
Mechanistic summaries	Yes	Uses simulated latent/ABM summaries; should be interpreted as an upper-bound-like mechanistic feature set in simulation.
Type I feature concatenation	Yes	Concatenates observed omics, spatial, and mechanistic features.
Type II residual correction	Yes	AI learns $Y - \hat{Y}_{MM}$ and adds correction to mechanistic prediction.
Type III AI-parameterized MM	Yes	AI infers hidden parameters from RNA/spatial features and predicts outcome through a mechanistic response-surface surrogate.
Type III+ parameterized MM + residual	Yes	AI-inferred parameters plus residual correction.
Type IV reduced PINN	Yes	PyTorch reduced state-space PINN with data loss plus ODE residual penalty.
Type V Neural-ODE-like residual dynamics	Yes	Learned residual vector field f_{NN} added to known reduced immune dynamics and rolled forward.
Type VII surrogate/emulator	Yes	Fast simulator surrogate; true-parameter versions are oracle diagnostics.
Bayesian hybrid	No	Intentionally excluded at this stage.
Full spatial PDE PINN	Not yet	Future version would learn D, I, C, R over (x, y, t) from sparse spatial data.
Fully differentiable ABM	Not yet	Future JAX/PyTorch implementation would backpropagate through virtual-cell interactions.

14 How real data can be fed into the model later

The simulated readouts are placeholders for real measurements.

14.1 Bulk RNA-seq

Real bulk RNA-seq can replace pseudo-bulk module features. Gene-set scores can be computed for TNF/NF- κ B, IL6/JAK/STAT3, IFN response, cytotoxicity, exhaustion, TGF- β , wound healing, hypoxia, apoptosis, macrophage, T cell, NK cell, Treg, and myeloid suppression programs. These become X_{bulk} .

14.2 Single-cell RNA-seq

Single-cell data can calibrate virtual-cell gene-program distributions:

$$\mathbf{g}_i(0) \sim p(\mathbf{g} \mid \text{cell type, patient, disease}). \quad (75)$$

It can also estimate transition rules between activation, exhaustion, cytotoxicity, regulatory, and repair states.

14.3 Spatial transcriptomics

Real spatial transcriptomics can replace simulated module maps $M_m(x, y)$. This enables direct computation of spatial exclusion, hotspot distances, core–rim contrasts, and colocalization features. If longitudinal spatial data are available, they can support Type IV PINN and Type V Neural-ODE-like dynamics more directly.

14.4 Imaging and histology

Histology, multiplex immunofluorescence, CODEX, imaging mass cytometry, or spatial proteomics can estimate cell density, immune access, barrier/fibrosis, and core–rim architecture. These can calibrate $B(x, y)$, $A(x, y)$, and cell-type spatial distributions.

14.5 Clinical outcomes and intervention response

Clinical response data can supervise the final outcome models:

$$Y_{\text{clinical}} = \text{response, progression, toxicity, relapse, survival, or biomarker decline}. \quad (76)$$

Intervention-specific data can train residual correction and few-shot drug adaptation.

15 Clinical and precision-medicine applications

The long-term use case is a patient-specific immune digital twin. Given patient omics, spatial data, imaging, and clinical variables, the model would infer latent immune-control parameters, simulate untreated progression, compare interventions, update predictions after early response, and estimate which treatment is most likely to reduce inflammatory burden while minimizing toxicity.

Potential applications include:

- inflammatory skin disease and tissue inflammation mapping;
- cancer immunotherapy response and immune exclusion;
- CAR-T/CAR-NK design and combination therapy optimization;
- autoimmune or chronic inflammatory disease stratification;
- cytokine-storm or hyperinflammatory syndrome modeling;
- drug repurposing and preclinical virtual trial design;
- patient-specific adaptive treatment scheduling.

16 Limitations

The current framework is a simulation benchmark. The RNA-seq and spatial maps are synthetic, not yet fitted to real cohorts. The mechanistic-summary arm is strong because latent states are known in simulation. The Type IV PINN is reduced-state, not a full spatial PDE PINN. The Type V Neural-ODE-like model is implemented as a discrete residual-vector-field learner, not a continuous differentiable ODE solver. Oracle true-parameter surrogate rows are diagnostic upper bounds. Real clinical translation will require calibration, validation, uncertainty quantification, independent cohorts, and prospective response data.

17 Recommended result interpretation

The main interpretation should not be that a single hybrid model always wins. Rather, different hybrid architectures answer different scientific questions. Feature concatenation asks whether mechanistic summaries add information. Residual correction asks whether AI can repair mechanistic bias. AI-parameterized MM asks whether patient-specific hidden biology can be inferred from data. PINN and Neural-ODE-like models ask whether learned dynamics can respect immune feedback laws. Surrogates ask whether expensive simulators can be replaced by fast predictors. Few-shot drug learning asks whether early treated examples can rapidly improve intervention prediction.

18 Conclusion

The updated model provides a broad non-Bayesian hybrid mechanistic–AI benchmark for inflammatory immune dynamics. It includes transcriptomics-informed virtual cells, spatial latent fields, RNAseq-like readouts, intervention counterfactuals, and multiple hybrid architectures, including residual correction, AI-parameterized mechanistic prediction, reduced PINNs, and Neural-ODE-like learned residual dynamics. The framework is a bridge from synthetic immune-tissue simulation toward real precision-medicine workflows in which patient-specific molecular, spatial, and clinical data inform interpretable, adaptive, and intervention-ready immune digital twins.